BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz - 2
Duration: 30 minutes

Semester: Summer 2025
Full Marks: 15

CSE 340: Computer Architecture

Name: Solution	ID:	Section:

1. X_{18} = 0x0000AE0000BCD011

→ Little endian

You are working with a 64-bit RISC-V machine. You want to store the contents of register X_{18} into memory, starting at address 10

a) Show how the data will be stored in your RISC-V machine. [4]

Answer:

Add.	Data
10	11
11	D0
12	BC
13	00

Add.	Data
14	00
15	AE
16	00
17	00

b) Complete the instruction below to carry out the operation described in question a. [2]

Addi X_{12} , X_0 , 10
SD X_{18} , 0(X_{12})

From → Reg.
To → mem] Store | 64 bit data - Double Word

2.

add x7, x28, x7 ✓ slli x5, x29, 1 → 2k slli x5, x5, 3 → 2k×8 add x5, x5, x11 → off+Base ld x28, 0(x5) → load B	Assume that the variables f, g, h, i, j and k are assigned to registers x5, x6, x7, x8, x28, and x29, respectively. Assume that A[0] and B[0] are in registers x10 and x11, respectively.

a) For the RISC-V assembly instructions above, what is the corresponding C/high level statement(s)? [5]

Answer:

h = j + h

j = B[2k]

b) What is the size of each index of array ~~A~~ B? [1]

Answer: 64 bits / 8 bytes.

3. Find the RISC-V code of the following machine Codes.
i. 0x**0392AAB3** **[3]**
Answer:

0 3 9 2 A A B 3

$$\underbrace{0000\ 001}_{f7}\underbrace{1\ 1001}_{rs2}\underbrace{0010\ 1}_{rs1}\underbrace{010}_{f3}\underbrace{1010\ 1}_{rd}\underbrace{011\ 0011}_{opcode}$$

opcode ↓ R type

AND X_{21}, X_5, X_{25}

Format	Instruction	Opcode	Funct3	Funct7/Funct6
R	ADD	0110011	001	0000000
	SUB	0110011	110	0000001
	AND	0110011	010	0000001
	OR	0110011	011	0000100
	SLL	0110011	101	0000100
I	LD	0000011	000	N/A
	LW	0000011	010	N/A
	LH	0000011	011	N/A
	LB	0000011	001	N/A
	ADDI	0010011	000	N/A
	SLLI	0010011	001	000000
	SRLI	0010011	010	000000
	ORI	0010011	011	N/A
S	SD	0100011	000	N/A
	SW	0100011	001	N/A
	SH	0100011	010	N/A
	SB	0100011	011	N/A